

Pablo Zúñiga Carvajal

Industrial & Systems Engineer — 6-Year Integrated Engineering Degree

Data Architecture • Applied Statistics • Complex Systems

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Viña del Mar, Chile • Open to Relocation • Visa Eligibility: US H-1B1 & EU Blue Card

Tech Stack

- **Data Engineering & Cloud:** Python (FastAPI, Polars), TypeScript, SQL, MS Azure, ETL/ELT pipelines, Docker, GitHub Actions (CI/CD), relational databases, and Parquet.
- **Advanced Analytics & Systems:** R (Tidyverse, targets), inferential & predictive statistical modeling, Monte Carlo simulations, BPMN/UML, data governance, and Docs-as-Code knowledge architectures.

Work Experience

- Cuxhaven Labs** (*Proprietary R&D Studio*) May 2026 – Present
Principal Systems Architect & Founder Santiago, Chile
- Architecting data systems and reproducible technical workflows through Git, Quarto, and Docs-as-Code.
 - Developing AstroGit-OS, a declarative architecture for astronomical instrumentation operational lineage.
- Systems Engineering & Advanced Analytics Advisory** Jul 2022 – Jul 2026
Independent Forward Deployed Engineering Consultant Valparaíso, Chile
- Engineered Python/R exponential statistical models optimizing GNSS pacing data across **1.8k+ triathlon athletes**.
 - Designed database structures, automated data flows, and Quarto/Markdown documentation architectures.
- Decentralized Systems & Web3 Protocols** Jan 2020 – Jul 2022
Web3 Systems & On-Chain Infrastructure Engineer Global / Remote
- Controlled DeFi assets via cold/non-custodial cross-chain wallets and DEXs networks.
 - Coordinated NFT smart contract deployments for **1,100+ concurrent users** in distributed environments.
- Pontificia Universidad Católica de Valparaíso** Aug 2018 – Jun 2021
Academic Data Engineer & Technical Consultant Valparaíso, Chile
- Engineered ETL pipelines for multi-source academic datasets; mentored STEM capstone & data science projects.
- Banco de Crédito e Inversiones (BCI)** (*Data & Analytics Corporate VP*) May 2019 – Sep 2019
Process Engineer Santiago, Chile
- Orchestrated metadata integration from core systems into MS Azure, standardizing **BPMN process workflows**.

Education

- Pontificia Universidad Católica de Valparaíso** 2011 – 2017
Industrial & Systems Engineering (Integrated 6-year program, BSc & MSc Equivalent) Valparaíso, Chile
- Core focus: systems theory, information technology, inferential statistics, operations research, and econometrics.

Projects & Internships

- ESVAL – Lead Capstone Project** | Viña del Mar, Chile Jul 2017 – Dec 2017
Spatio-temporal predictive risk model: Cox Proportional Hazards in R (5,300 km of pipeline 1.9M+ residents).
- ENAMI – Industrial Engineering Intern** | Atacama & Santiago, Chile Jan 2016 – Mar 2016
Audited metallurgical inventory database (75k+ SKUs), uncovering \$2.5M USD in accounting discrepancies.
- Antofagasta Minerals – Geotechnical Engineering Intern** | Salamanca, Chile Jan 2014 – Mar 2014
Monitored open-pit mining geotechnical telemetry for slope failure prediction and early risk mitigation.

Technical Instruction

- Pontificia Universidad Católica de Valparaíso – School of Business** 2019
Graduate Technical Instructor Valparaíso, Chile
- BancoEstado / Universidad Federico Santa María** 2017, 2019
Advanced Analytics Speaker & Trainer Santiago, Chile
- Pontificia Universidad Católica de Valparaíso – School of Industrial Engineering** 2018
Guest Lecturer (Applied Data Science) Valparaíso, Chile